

Elvis F. Elli

From: rebekah.hanson@usda.gov
Sent: Thursday, August 20, 2026 1:55 PM
To: Elvis F. Elli
Subject: [EXTERNAL] Review Package for Proposal 2025-11488 Submitted to USDA\NIFA

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Program: Food and Agriculture Cyberinformatics and Tools

Proposal Number: 2025-11488

Project Director: Elvis Elli

Proposal Title: Integrated data-driven approaches to advance double-cropping recommendations for maximizing yields and soil health in the US Mid-South

Dr. Elvis Elli

Division of Agriculture of the University of Arkansas System

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Fayetteville, AR 72704-6898

Dear Dr. Elli,

Thank you for submitting your application to NIFA's Agriculture and Food Research Initiative A1541 Data Science for Food and Agriculture Systems (DSFAS) program. A carefully selected panel of experts reviewed your proposal, and we regret to inform you that your application was not recommended for funding in this round of competition. This does not mean that your proposal lacked merit. A careful examination of the attached reviews should provide a general indication of whether resubmission is warranted.

DSFAS received a large number of very-high-quality applications and the bar was set high for being able to attain the expectations of the program. We encourage any unsuccessful applicant to consider the comments of the peer panel and reapply.

We wish you the best in future applications.

Regards,
NIFA-DSFAS@usda.gov

Steven Thomson, Ph.D

National Program Leader

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United States Department of Agriculture
National Institute of Food and Agriculture
Agriculture and Food Research Initiative
Food and Agriculture Cyberinformatics and Tools

Proposal 2025-11488 submitted by Elli

Integrated data-driven approaches to advance double-cropping recommendations for maximizing yields and soil health in the US Mid-South

The review panel grouped proposals into one of the relative categories below. The percentage indicates the final distribution of proposals in each category.

Outstanding %	2
High Priority %	12
Medium Priority %	62
Low Priority %	21
Do Not Fund %	3

This proposal was placed in : Medium Priority

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Food and Agriculture Cyberinformatics and Tools - PANEL SUMMARY

The panel decision regarding your proposal is based on the input provided by the reviews and the collected expertise and judgment of the individual panel members. This panel summary reflects the consensus opinion of the panel regarding your proposal.

Proposal Number: 2025-11488 Project Director: Elli

Proposal Title: Integrated data-driven approaches to advance double-cropping recommendations for maximizing yields and soil health in the US Mid-South

Positive Aspects of the Proposal

The panel agreed that the proposed project clearly justifies the need to study the double cropping system. The proposed methodology, which leverages field observations, remote sensing, and process-based modeling, provides a lot of detail and aligns well with the program's priorities. The application of process-based modeling for scenario analysis associated with the proposed treatments and system can help bridge existing research and data gaps in this data scarce system. The team is well qualified for the proposed work, and the proposal is generally well organized.

Negative Aspects of the Proposal

The panel is concerned that the study focuses on a single site and includes a limited number of crop phases that can be sampled within the project. This scope would pose a significant challenge for scaling the results to the Mid-South and for accounting for the variability in outcomes related to weather fluctuations. The panel also notes a lack of detail regarding the R Shiny tool and how stakeholder feedback will be incorporated to achieve real-world impact. It was also pointed out more supporting letters from stakeholders are needed.

Synthesis Comments

Overall, the proposal has some good potential to advance agricultural management in double cropping systems in the U.S. Mid-South. The proposal presents the prototype monitoring system clearly for a selected site, but it lacks sufficient sample size in both space and time to ensure adequate coverage of the target system. It is a strength that the proposal considers stakeholder engagement for communicating model outcomes; however, this component requires further development to clarify how stakeholders will be involved and how they will benefit from the proposed interface.

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The following reviews were submitted for your proposal, the names of the reviewers have been removed to maintain confidentiality.

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Short Overview:

Aims to improve adoption of double-crop grain systems in US midsouth region.

All field research sites focused on soybean production systems in AR.

Scientific Merit of the Application for Research:

Effectively introduces the idea that double crop wheat-soybean systems have significantly higher yield, protein production, and water use efficiency in this region.

Considers emerging winter crops like canola and oats.

Identifies soybean planting date pressure as reason for reduced wheat acreage; does not address changes in wheat costs of production or minimal yield gains.

APSIM has been used for double cropping systems, but not extensively. Preliminary data shown for one region of wheat-soybean systems.

Objective #1. Conduct field-scale experiments testing multiple cropping sequences to enhance understanding of soil-plant mechanisms driving crop yields and soil health

The 8 treatments make sense for modeling and extension purposes, but it's not clear if farmers using the winter wheat based treatments would use full season corn at any point in these rotations.

Single location

Soil health sampling plan is well rounded.

Objective #2. Use multispectral data at multiple scales to identify key metrics across double cropping systems for improved biomass production and seed yields

Hyperspectral imagery has been used on all listed crops in past research, indicating feasibility.

Objective #3. Identify double-cropping options to maximize yield levels and stability for target environments using a crop growth model (APSIM)

Models will compare crop yield, yield sensitivity, and protein along with WUE across 1985 through 2024; captures carryover effects.

R Shiny interface is included in this objective; will allow systems comparisons and visualize optimal management. Unclear what management practices will be assumed, which may impact usability, and does not describe a user experience study.

Sensitivity analysis for isolating impacts of previous crops is a helpful addition to the study (see pitfalls section)

Qualifications of Project Personnel, Adequacy of Facilities, and Project Management:

Team has appropriate modeling, imagery, and field research experience.

Institution has appropriate equipment and computing resources.

Sustainability plan of shiny website does not include specific timeline for access after the end of the project, but data will be available in public repositories.

Project Relevance:

Applies to AST and PHPP priority areas.

Has extensive data science applications; does not appear to include AI in modeling plans, which makes sense given the volume of data described for field research activities.

Other Notes:

Overall, it's wise to integrate emerging crop research within systems with crops with known markets and profitability; allows adoption with less risk.

Fewer letters of support than many other proposals.

Mentoring plan is brief but includes key features like IDPs and career planning.

Overall Comments

This proposal seems to provide more scientific value for production agriculture sciences than for data sciences as the data science element of the proposal seems to lack novelty. Considering the applied nature of this proposal, letter of supports are lacking from key industry partners (ie Arkansas Soybean Producers Association). A R-Shiny application is mentioned to be developed, but functionality and stakeholder engagement in the tool have limited discussion.

Strengths

The proposal address a need to provide more robust data-driven information to producers interested in diversifying their cropping systems

Some approaches for objectives are described in detail

Preliminary data has been collected on the modeling side of the project although it is a bit unclear how the preliminary efforts connect to the proposed work

Weaknesses

There is only one letter of support provided from an organization in California that manages farms in the Mid-South USA. It would be stronger to have support from the soybean and small grain growers of Arkansas

The developed R-Shiny app should be more thoroughly discussed in dissemination of results section

Sustainability of the web-based decision support tool is weakly described

In the timeline, double crop systems of investigation (Objective 1) would appear to only have one cycle (2 at max) they could be investigated which leaves results very prone to impact by weather variability (the same unpredictable weather variability discussed in this proposal!). Additionally it appears field work will only be conducted in one location since so the field element of the proposal is narrow in scope to drive subsequent data science efforts

The study timeline is relatively short to be assessing impacts on soil health

Limitations for data collection (ie cloud cover) to drive objective 2 (Identify key hyperspectral differences across double-cropping systems) are discussed and underscore the need for this work to be conducted at great than 1 sites for 2 years

The development of the R-Shiny web interfaced is under-discussed throughout the proposal and the proposal lacks efforts to have stakeholder co-develop and beta-test this tool to fit their needs

There is lacking information on how to information generated from this project will be used to inform agricultural practitioners on improved double crop decision making

Overall

o Strengths

This project aims to combine field experiments, remote sensing data analysis, and process-based modeling to improve decision making for double-cropping practices in the U.S. Mid-South. The project addresses important data and modeling gaps for a region that urgently requires improved management but has received limited prior research attention. The work also has potential for real-world impact through model-enabled scenario analysis incorporated into the web interface for extension activities. The team is well qualified to conduct this work, and the preliminary modeling results provide good support for the feasibility of the proposed activities. The justification, methodology, limitations and mitigation plans, and expected outcomes are described clearly.

o Weaknesses

The proposal could benefit from additional detail on how the proposed site-level study will eventually scale up to support decision making for double-cropping systems across the U.S. Mid-South. In addition, the proposal could explain more explicitly how the PI and Co-PIs will collaborate throughout the project. Further clarification about how the R-Shiny web interface will engage stakeholders and incorporate their feedback would also strengthen the proposal.

Resubmission (if applicable)

o Comments

Not applicable.

Scientific Merit

o Strengths

The proposal clearly describes what crop physiological and soil health measurements will be collected across space and time, which treatments will be monitored, what additional variables (e.g., gas exchange) will be measured, which satellite sensors will be utilized, and how APSIM will be calibrated and applied for scenario analysis. While the methods themselves are widely used and well established, this study aims to leverage a complex monitoring system in a data-scarce environment.

o Weaknesses

There are no major weaknesses identified, except that a site-level study spanning only a few years of monitoring cannot fully capture the complexity of interactive weather, soil, and management effects across the U.S. Mid-South.

Qualifications

o Project Director

PI is well-qualified to lead this project.

o Co-project Director(s)

Co-PIs are well qualified to support this project.

o Other key personnel

The project will provide training opportunities for a graduate student and a postdoc, which is a clear strength. However, it is not clear who will develop the R shiny App from the key personnel and budget justification statements.

Project Relevance

o Strengths

The project is relevant to DSFAS due to its interdisciplinary nature. In particular, this project is relevant to NIFA priorities related to Agriculture Systems and Technology and Plant Health and Production and Plant Products.

o Weaknesses

No weaknesses.

Data Management Plan

o Strengths

The management plan clearly states the anticipated data type, format, and sharing plans.

o Weaknesses

No weaknesses.

Budget and Period of Support

o Comments

The budget is reasonable for the proposed scope of work.